

Components Study

Arduino UNO

Arduino UNO is a popular microcontroller development board based on the ATmega328P microcontroller. It is widely used for learning embedded systems, interfacing sensors, and developing automation projects.

Features:

- 14 Digital I/O Pins
- 6 Analog Input Pins
- USB Programming Support
- Easy to use and low cost

ESP32

ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth connectivity. It is commonly used in IoT applications and wireless communication projects.

Features:

- Built-in Wi-Fi and Bluetooth
- Dual-core processor
- Low power consumption
- Supports IoT applications

Raspberry Pi Pico

Raspberry Pi Pico is a low-cost microcontroller board based on the RP2040 chip. It is suitable for embedded systems, robotics, and sensor-based projects.

Features:

- Dual-core ARM processor
- Multiple GPIO pins

- Low cost and compact size
- Supports MicroPython and C/C++

Temperature Sensor

A temperature sensor measures the temperature of the surrounding environment and converts it into an electrical signal that can be processed by a microcontroller.

Applications:

- Weather monitoring
- Industrial automation
- Medical devices

Motion Sensor (PIR Sensor)

A PIR (Passive Infrared) sensor detects motion by sensing changes in infrared radiation emitted by humans or animals.

Applications:

- Security systems
- Automatic doors
- Smart lighting

Light Sensor (LDR)

An LDR (Light Dependent Resistor) detects the intensity of light. Its resistance changes according to the amount of light falling on it.

Applications:

- Automatic street lights
- Smart lighting systems
- Light intensity monitoring